ANALYSTLAB AFRICA

Week 6 Data Quality Report

HealthConnect Rwanda - Healthcare Predictive Analytics and Population Health Intelligence

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Synthetic educational healthcare data. Analytical outputs are not clinical advice or diagnosis.

1. Executive finding

The supplied package contains 25 CSV files plus a six-sheet Clinical Transactions workbook. The detailed appointment sheet contains 5,000 appointments, while full diagnosis and billing tables extend to 60,000 AppointmentIDs. The enterprise revenue summary reports 60,809 annual appointments. This means the 5,000 detailed appointments are treated as an analytical sample for appointment-level modelling, while enterprise forecasting uses the monthly summary.

2. Dataset quality profile

Dataset	Rows	Cols	Missing	Duplicates
Clinical_Admissions	5,000	3	0	0
Clinical_Appointments	5,000	7	0	0
Clinical_Diagnoses	5,000	4	0	0
Clinical_Laboratory_Tests	5,000	4	0	0
Clinical_Prescriptions	5,000	4	0	0
Clinical_Referrals	4,708	3	0	0
HCCountries	1	6	0	0
HC_Administrative_Staff	80	10	0	0
HC_Beds	2,000	3	0	0
HC_Billing	60,000	4	0	0
HC_Departments	200	3	0	0
HC_Diagnoses	60,000	4	0	0
HC_Districts	30	3	0	0
HC_Doctor	350	10	0	0
HC_Medical_Equipment	200	4	0	0
HC_Medical_Specialties	200	3	0	0
HC_Nurse	700	10	0	0
HC_Ownership_Types	4	2	0	0
HC_Patient_Feedback	21,064	4	0	0
HC_Patients	20,000	19	5,071	0
HC_Payments	60,000	4	0	0
HC_Provinces	5	3	0	0
HC_Radiographer	30	10	0	0
HC_Revenue_Summary	12	3	0	0
HC_Rooms	400	4	0	0
HC_Sectors	120	3	0	0
HC_Services	400	3	0	0
HC_Villages	480	3	0	0
Insurance_Claims	42,036	5	0	0
Laboratory_Scientist	40	10	0	0
Pharmacist	50	10	0	0

3. Relationship validation

Relationship	Unmatched	Match rate
Patients.VillageID -> Villages.VillageID	0	100.0%
Appointments.PatientID -> Patients.PatientID	0	100.0%
Appointments.DoctorID -> Doctors.StaffID	0	100.0%
Appointments.DepartmentID -> Departments.DepartmentID	0	100.0%
Diagnoses.AppointmentID -> detailed Appointments	55,000	8.3%
Billing.AppointmentID -> detailed Appointments	55,000	8.3%
Feedback.AppointmentID -> detailed Appointments	19,308	8.3%
Beds.RoomID -> Rooms.RoomID	0	100.0%
Rooms.DepartmentID -> Departments.DepartmentID	0	100.0%
Equipment.DepartmentID -> Departments.DepartmentID	0	100.0%

4. Material data limitations

Health Facilities master missing: FacilityID values HF001-HF020 are used throughout the data, but no supplied table defines facility names, types, ownership or geographic links. Facility reporting therefore remains anonymised by FacilityID.

Department-Facility hierarchy inconsistency: only 5.1% of appointment DepartmentID-FacilityID pairs match the Department master; doctors and nurses show similarly low consistency, while Rooms and Medical Equipment are fully consistent. DepartmentID is therefore used for department labels, but the Department master is not treated as a reliable cross-facility hierarchy for appointments/staff.

Insurance coverage comparison requested in the brief cannot be performed as insured vs uninsured because every patient record contains an InsuranceID. Alternative supported statistical comparisons are used instead.

All 2,000 supplied beds are marked Available. This variable therefore has no variation and cannot support meaningful bed-occupancy prediction or facility capacity ranking.

5. Responsible-use decision

No missing facility names, ownerships or geographic facility mappings are fabricated. The HighRiskPatientIndicator is a transparent educational heuristic for feature-engineering demonstration only; it must not be used for clinical triage or care decisions.

6. Analytical relationship map

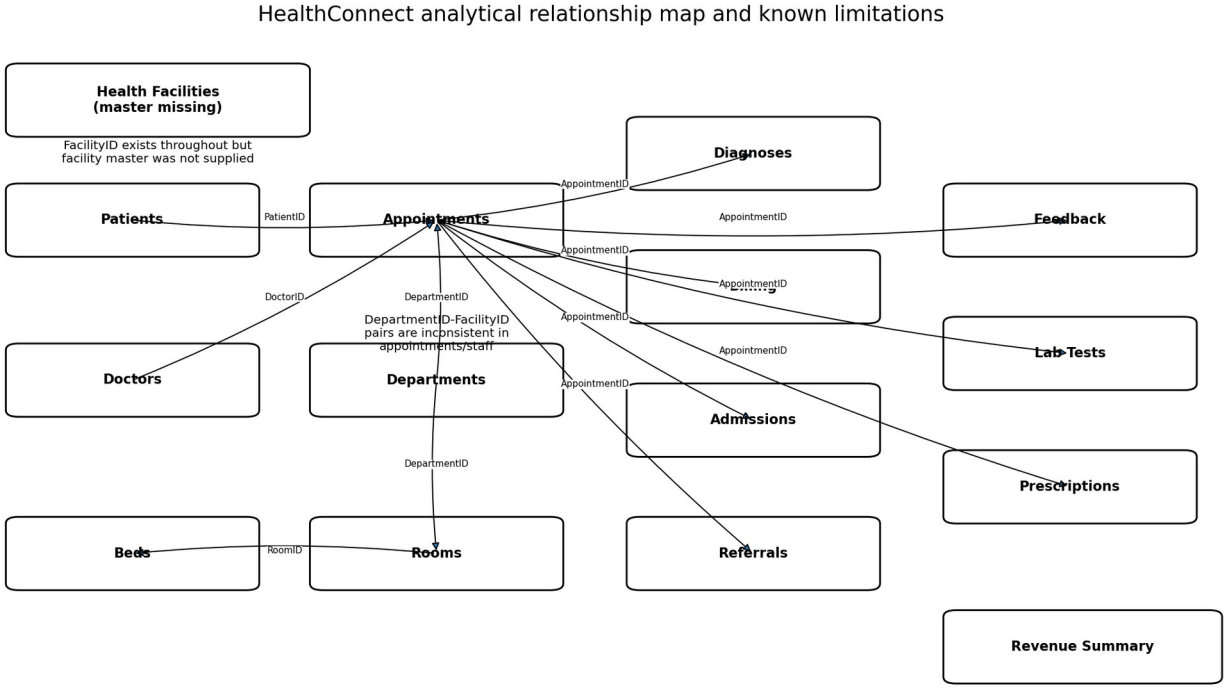


Figure 1. Validated analytical relationships and known master-data limitations.